

Continual test-time adaptation via weight averaging of feature augmentations in cross-domain medical image segmentation

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ABSTRACT

Continual Test-Time Adaptation (CTTA) effectively mitigates domain shifts that occur across medical institutions by allowing models to adapt at test time without revisiting the source data. Existing CTTA methods exploit transform invariance through pixel-level augmentation, but these approaches tend to produce inconsistent results in online environments, particularly critical in semantic segmentation tasks. To address these issues, we propose FAWA, a CTTA framework that combines feature-based augmentation and a weighted averaging strategy. FAWA augments features by retrieving statistical information from a continuously updated memory bank, which generates diverse perturbations of input target samples. Each transformed sample is independently fine-tuned, and the resulting parameter sets are merged through continuous weighted averaging. This process leads the model to converge to a stable minimum and mitigates the accumulated error. Experiments on Brain CT and Prostate MRI benchmarks indicate that FAWA achieves competitive performance while ensuring robustness during extended adaptation streams.

1. Introduction

Semantic segmentation of organs remains a fundamental task in medical image analysis, allowing precise delineation and quantification of anatomical structures. However, most segmentation models experience significant performance degradation when applied to unseen domains. This degradation is primarily caused by the domain shift, which is based on variations in scanner vendors, acquisition protocols, noise levels, brightness, and contrast, leading to institution-specific imaging characteristics. Although domain adaptation (DA) methods (Zou et al., 2018; Guan and Liu, 2021; Liu et al., 2025) have been extensively studied to address this issue, they typically rely on access to labeled or unlabeled data from the target domain. In real-world clinical scenarios, however, access to such data is often infeasible, as only pre-trained model weights are available, while raw source data are restricted. Moreover, collecting additional training set from unseen domains is generally impractical due to data scarcity or privacy issues. Given these constraints, Continual Test-Time Adaptation (CTTA) has emerged as a promising alternative. CTTA aims to continuously adapt a model to incoming test samples in real time without access to source data or retraining from scratch. Recent CTTA methods commonly exploit transform invariance, which assumes that a model's predictions should

remain consistent across various transformations of the input. Based on this idea, many methods apply diverse image augmentations to simulate domain shifts and guide model adaptation. Specifically, M. Zhang et al. (2022) minimize entropy loss across multiple augmented views to encourage confident and stable predictions, while Wang et al. (2022) generate pseudo-labels by filtering predictions based on confidence scores. Prediction consistency between original and augmented inputs (Chen et al., 2025, 2022) is also frequently used to ensure robustness. Pixel-wise transformation schemes, however, frequently struggle to maintain consistent performance within online CTTA environments. Ill-suited or excessively aggressive augmentations risk compromising a medical image's structural integrity and semantic consistency. Such distortions invariably lead to unreliable predictions, the accumulation of errors, and, in severe instances, even model collapse (Nguyen et al., 2023). Prior research including works by Nguyen et al. (2023), M. Zhang et al. (2022), has underscored how performance can fluctuate based on the chosen augmentation strategy, especially in fine-grained tasks where intricate transformations are prone to causing information loss (Ma, 2024). While this inherent fragility is well documented within classification contexts, its effects are amplified in semantic segmentation, where the precise preservation of spatial accuracy is paramount.

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To overcome these shortcomings, we propose FAWA, a CTTA framework that combines Feature Augmentation with Weighted Averaging. FAWA includes (1) Top-K Feature Augmentations, which mixes higher-order feature statistics retrieved from a memory bank, and (2) fuses the individually fine-tuned weights via Continuous Weight Averaging, which pulls the model toward a common minimum and arrests error accumulation. Unlike earlier feature-space perturbations that inject random noise (Li et al., 2021), this memory-guided mixing preserves the spatial structure of the target representation while exposing the model to diverse statistical variations drawn from previously observed target samples. Each augmented variant is then independently fine-tuned, and the resulting parameters are merged via Continuous Weight Averaging, pulling the model toward a common minimum and suppressing error accumulation. Fig. 1 illustrates these dynamics on the two-dimensional projection of the loss landscape. Beginning from the pre-trained point PT, the model adapted using the proposed FAWA method follows a trajectory that gradually converges to points R1, R2, and R3 over three adaptation rounds. At the end of each adaptation round, three independently fine-tuned models, FA₁, FA₂, and FA₃ are obtained, each trained using three distinct feature augmentations. While these three represent a subset, the full FAWA process utilizes ten different feature augmentations in total. Since all three lie within a shared flat basin on the loss landscape, averaging their weights pulls the parameters toward the center of this region, resulting in a combined model FA_{avg} being positioned near R1. The red-shaded region visualizes the standard deviation of FAWA trajectories across adaptation rounds, reflecting the consistency of the learning dynamics. While conventional pixel-level augmentation techniques, such as 3D flipping or the combined operations in SimAug (random flipping, rotation, and cropping), show similar general trends during adaptation, they frequently drive models towards suboptimal local minima along shorter adaptation paths. A key drawback of these traditional methods is their inability to ensure stable predictions or to reliably generate diverse feature representations that effectively converge towards the target distribution. This often results in suboptimal adaptation performance. Our proposed FAWA framework addresses this by employing feature-based augmentation, which actively guides the adaptation process. By sampling diverse representations that are well aligned with the target distribution, FAWA prevents premature convergence. Moreover, the weighted average of parameters across these augmented variants promotes improved local stability, which in turn leads to more robust generalization and consistent adaptation ability when confronted with unseen datasets. Our key contributions are summarized as follows:

- Demonstrate that pixel-level augmentations amplify errors during online CTTA for medical image segmentation, whereas the proposed feature-space perturbations mitigate this failure mode.
- Propose FAWA, which augments features by mixing statistics retrieved from a memory bank and fuses adaptation weights via continual weight averaging, enabling stable one-pass adaptation from a single input sample.
- Experimental results for brain CT and prostate MRI benchmarks indicate that FAWA demonstrates competitive performance while ensuring stability during long adaptation streams.

2. Related work

2.1. Continual test-time adaptation

CTTA enables online, single-pass adaptation in settings where only a pre-trained model is available. Early approaches such as TENT (Wang et al., 2021) and its sharpness-aware variant SAR (Niu et al., 2023) improve performance by minimizing prediction entropy. A complementary line of work stabilizes predictions through consistency across augmented views (Wang et al., 2022; Chen et al., 2022, 2025). CTTA

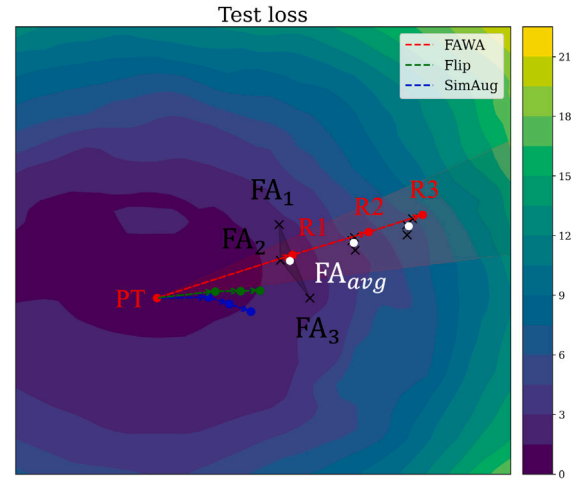


Fig. 1. Test loss landscape of the model pre-trained (PT) on the GLIS-RT dataset, with projections of adapted models visualized following Li et al. (2018). The red dashed arrow represent the model adapted using FAWA, while the green and blue dashed arrows represent models adapted with flip and SimAug, respectively. The projection is performed on the loss surface centered at PT using an orthonormal basis defined by directions from PT to models adapted with different augmentations, enabling visualization of adaptation trajectories in a shared 2D subspace.

has since expanded beyond classification to detection and segmentation (Cao et al., 2023; Liu et al., 2024), with medical image segmentation emerging as a major application domain (Yang et al., 2022; Chen et al., 2024; Zhang et al., 2024). Recent extensions also cover multi-modal backbones. For instance, WATT (Osowiechi et al., 2024) adapts CLIP prompt embeddings via weight averaging, a strategy similar to our proposed FAWA method. By operating at the prompt level, Chen et al. (2024) decouples adaptation from backbone parameter updates, thereby reducing catastrophic forgetting.

2.2. Using feature statistics in domain shift problems

Domain Generalization (DG) mitigates domain shift by learning representations that transfer to unseen domains. Although image-level augmentation is a core DG practice, its task-specific nature limits diversity. Recent studies therefore exploit the statistics of intermediate features, primarily means and variances, to synthesize novel domain variations (Li et al., 2021; Huang and Belongie, 2017; Zhou et al., 2024; Y. Zhang et al., 2022). AdaIN (Huang and Belongie, 2017) first aligned features via mean-variance matching, later work stochastically mixed statistics across domains with MixStyle (Zhou et al., 2024). Moving beyond Gaussian assumptions, Exact Feature Distribution Matching (Y. Zhang et al., 2022) aligns full empirical cumulative distribution functions, better capturing the skewed activations produced by ReLU. Collectively, these statistic-based feature augmentations enrich representational diversity and markedly strengthen DG. Such feature-level strategies have also been applied to medical image segmentation, where domain shifts across scanners and protocols are pronounced (Jiang et al., 2024).

2.3. Weight ensembles

Weight ensembling likewise enhances generalization. Stochastic Weight Averaging (SWA) averages checkpoints along the optimization trajectory, guiding the model toward flatter minima and lower loss values (Izmailov et al., 2018). Empirical evidence shows that models fine-tuned from a shared pre-trained seed converge to the same loss basin. Consequently, linearly interpolating their weights preserves,

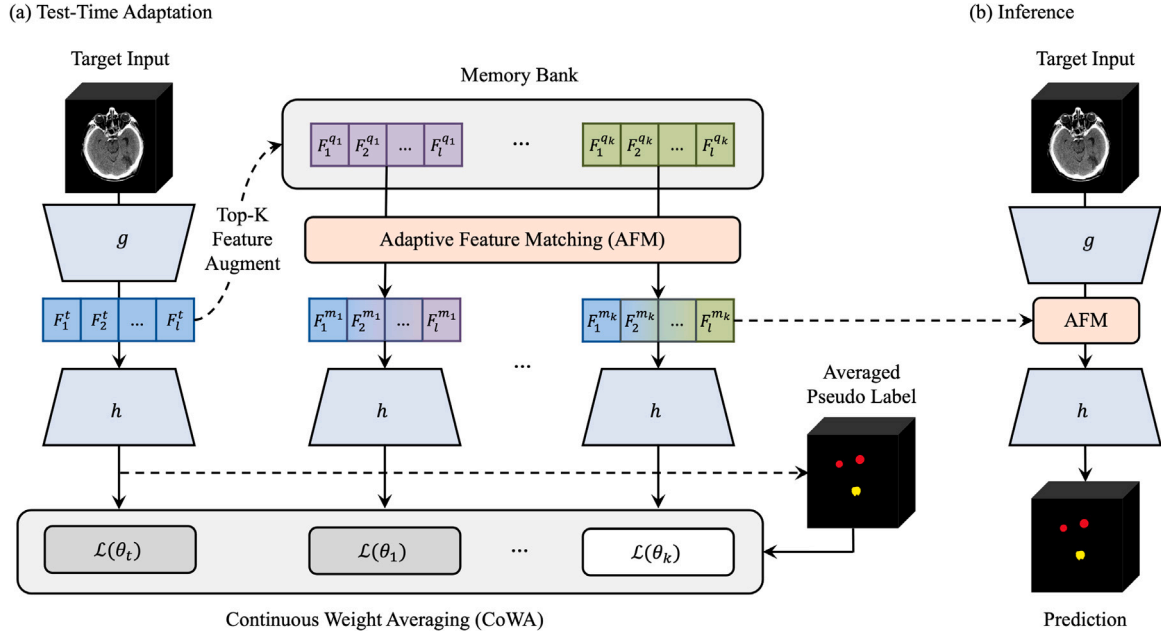


Fig. 2. Overview of the proposed FAWA framework. In (a), the model is adapted using multiple augmented features, and the one with the highest Dice similarity to the pseudo-label (i.e., F^{m_k}) is selected. This feature is reused in (b), where AFM is applied during the inference phase. The final prediction is generated using the updated model. Solid lines indicate forward-pass paths, and dashed lines represent intermediate computations.

or even improves performance (Rame et al., 2022). Building on this insight, later work averages multiple independently fine-tuned models, further boosting robustness and generalization (Wortsman et al., 2022; Marouf et al., 2024). Notably, Continual Model Averaging (Marouf et al., 2024) adapts this strategy to continual learning by sequentially averaging weights, balancing plasticity and stability and thus mitigating catastrophic forgetting.

While feature statistics-based augmentation and weight averaging have been studied separately, their joint application in a continual TTA setting is non-trivial. Weight averaging is effective when independently fine-tuned solutions converge to neighboring regions in parameter space, a condition that depends critically on how augmentations are generated. Augmentations that are too diverse or misaligned with the target distribution can scatter solutions across the loss landscape, undermining the averaging operation. To our knowledge, this interdependency has not been explicitly addressed in prior work, where the two strategies are applied in isolation. FAWA is designed around this constraint, treating augmentation generation and weight averaging as jointly dependent components rather than independent modules.

3. Methodology

3.1. Overview

Let the labeled source domain be $D^s = \{(\mathbf{x}_i^s, \mathbf{y}_i^s)\}_{i=1}^{N^s}$, and the unlabeled target domain be $D^t = \{\mathbf{x}_i^t\}_{i=1}^{N^t}$. Here, each input $\mathbf{x}_i \in \mathbb{R}^{H \times W \times D}$ represents a 3D volume, and $\mathbf{y}_i^s$ denotes the corresponding ground-truth label, which is available exclusively in the source domain. Because the target distribution drifts over time ($D^s \neq D^t$), the model must adapt continually. A pre-trained model $f = g \circ h$, with encoder g and a segmentation head h , is initially trained on D^s and has parameters $\theta = \{\theta_g, \theta_h\}$. CTTA updates θ during deployment to mitigate performance loss on D^t . We propose FAWA, a continual test-time adaptation framework for medical image segmentation that combines two ideas in a single online loop. First, Top-K feature augmentation (FA) retrieves the K most similar feature maps from a memory bank and mixes their higher-order statistics with those of the current sample, producing several diverse yet spatially consistent feature variants. Next, the model

is fine-tuned independently on each variant, after which continuous weight averaging (CoWA) merges the resulting parameter sets. The averaged weights guide the network to converge toward a common minimum, leading the model to a balanced point among the different FA solutions, as shown in Fig. 1. This result is used as the initialization for the next target sample, effectively suppressing error accumulation as the target distribution changes. The overall architecture of the proposed framework is shown in Fig. 2. Given a target sample $\mathbf{x}^t$, the encoder produces F^t , which is augmented through FA to generate K variants of mixed features, denoted as $\{F^{m_1}, \dots, F^{m_K}\}$. Both the original and mixed features are passed through h , resulting in $K + 1$ logit maps, whose average is used as the pseudo-label. The segmentation loss updates only the affine parameters of the normalization layers, yielding a set of parameters $\{\theta'_0, \dots, \theta'_K\}$. CoWA then averages these parameters to derive the final weights θ_{avg} , which are used to initialize the model for the next target sample, enabling robust adaptation throughout stream.

3.2. Top-K feature augmentation

Top-K Feature Augmentation (FA) operates in feature space by applying statistical perturbations to enhance model generalization, while preserving the raw input. Specifically, FA targets latent features extracted from intermediate encoder layers, generating diverse and semantically meaningful variations by effectively leveraging feature statistics accumulated from earlier samples during test time. The FA process comprises two core components, adaptive feature matching and a memory bank.

3.2.1. Adaptive feature matching

Adaptive Feature Matching (AFM) improves alignment accuracy within feature space by capturing higher-order moments beyond traditional mean and variance statistics. Unlike global statistics-based methods (Huang and Belongie, 2017; Zhou et al., 2024), which collapse spatially heterogeneous feature maps into scalar summaries, AFM employs full empirical distribution matching to preserve the fine-grained distributional structure critical for segmentation. Given the constraints of TTA setting where source domain data is inaccessible, AFM retrieves query vectors with the highest cosine similarity from a memory bank

to the current target features, which stores feature representations obtained from previous target input streams. Feature vectors obtained from intermediate encoder layers in segmentation models encode rich spatial semantics (Chung et al., 2022), and previous work has demonstrated that shape similarity strongly correlates with transfer performance across domains (Zhang et al., 2024). Leveraging this insight, similarity-based query retrieval facilitates the generation of augmented feature combinations that are structurally coherent. Formally, given a target image volume $\mathbf{x}^t \in \mathbb{R}^{H \times W \times D}$, the latent feature extracted from the i th layer of encoder g is denoted by $F^i \in \mathbb{R}^{C_i \times H_i \times W_i \times D_i}$. The extracted target vector F^t and the query vector F^q are mixed by aligning their empirical distributions through rank-ordered sorting (Coltuc et al., 2006; Y. Zhang et al., 2022), as follows:

$$F^m = \Pi_{F^t}^{-1}((1 - \lambda)\Pi(F^t) + \lambda\Pi(F^q)), \quad (1)$$

where $\Pi(\cdot)$ represents the sort-matching function (Sedgewick, 1978), and the mixing coefficient $\lambda \in [0, 1]$ controls the contribution of each distribution to the resulting mixed vector F^m . The function $\Pi(\cdot)$ transforms the features into a rank-ordered representation that is independent of spatial positional information by sorting the values in ascending order. This mapping allows F^t and F^q to undergo direct numerical mixing on the same probabilistic rank, regardless of their respective spatial architectures. Subsequently, the mixed values are redistributed to the original indices of F^t through the inverse process. This generates a F^m that preserves the original spatial semantics while selectively adapting the statistical characteristics of the target distribution. This operation is applied to the intermediate layers l of the U-Net-based segmentation model, including the final encoder layer and skip connections (e.g., blocks 1–4 of ResUNet-50). The mixing coefficient λ is dynamically determined for each sample based on the average cosine similarity between the target and the query vectors as,

$$s = \frac{1}{N} \sum_{n=1}^N \frac{f^q \cdot f^t}{\|f^q\| \|f^t\|}. \quad (2)$$

The similarity score s is min-max normalized to the range $[0, 1]$ to obtain λ . A higher λ , assigned to queries with greater similarity, allows a stronger distributional influence from F^q , while dissimilar queries contribute minimally, preserving the statistical characteristics of the original target features. This mixing-based feature perturbation enables the generation of various statistical combinations from a single sample and serves as an effective method of feature augmentation.

3.2.2. Memory bank retrieval

AFM can be enhanced by employing a top-k retrieval approach, which allows the generation of multiple perturbed feature variants from a single input instance. In this context, the memory bank serves as a core component and is continuously updated during the TTA process. Updates are managed using a Least Frequently Used (LFU) replacement policy, where newly created mixed vectors from the EFDM are added as potential query entries for subsequent retrieval. The memory bank initially operates under a cold-start condition, where it is empty and incrementally populated as the TTA process proceeds. Once memory exceeds its storage limit, vectors with the lowest access frequency are discarded. The choice of LFU over alternative replacement policies (e.g., FIFO, Random) was determined empirically, thereby ensuring that frequently retrieved entries, whose statistics consistently contribute to the alignment of subsequent samples, are preserved as high-utility queries and supporting balanced adaptation under continual distribution shifts. Additionally, when used in conjunction with the CoWA strategy introduced in Section 3.3, the memory bank also acts as a temporal cache of query statistics, facilitating the reuse of augmentation patterns that were previously optimized.

3.3. Continuous weight averaging

TTA is known to suffer from error accumulation when adapting to environments with small batch sizes and frequent distribution shifts. This arises as the model overfit to specific inputs or amplifies incorrect predictions due to biased loss signals. As a result, accumulated errors degrade performance over time and may ultimately cause model collapse. To address this issue, CoWA applies weight averaging over successive fine-tuning steps. This produces a consolidated set of parameters that retains generalizability throughout the adaptation. As prior work suggests (Marouf et al., 2024), fine-tuning from a shared pre-trained initialization tends to guide optimization within a consistent region of the loss landscape, a property that remains under TTA scenarios. Memory-based FA introduces representational diversity by exposing the model to multiple, slightly perturbed views of the target domain. This diversity leads to multiple local optima that are closely located within the loss landscape. The integration of FA with a shared initialization guides fine-tuning instances along coherent optimization trajectories that converge toward neighboring regions in parameter space. Empirical support for this phenomenon is provided in Fig. 1, where the adapted solutions corresponding to FA_1 , FA_2 , and FA_3 are shown to converge towards nearby regions in the parameter space as the adaptation progresses from R1 to R3. While each adapted model captures distinct variations, their resulting solutions remain anchored within a shared region. CoWA leverages this by aggregating the locally adapted weights, thereby facilitating convergence toward a more stable and representative parameter configuration that better reflects the underlying target distribution.

3.3.1. Model adaptive strategy

FA generates K augmented feature maps that, together with the original target feature map, form $K + 1$ parallel branches in CoWA. During each forward pass, the original vector F^t (branch $k = 0$) and the K mixed vectors $\{F_1^m, \dots, F_K^m\}$ are processed by the segmentation head h , producing $K + 1$ logit maps. Averaging these logits yields a pseudo-label that supervises adaptation. For every branch, a standard segmentation loss (e.g. cross entropy or Dice) is computed, and the network is fine-tuned over L iterations, producing an updated parameter set $\{\theta'_0, \dots, \theta'_K\}$. These parameters are finally averaged to obtain the adapted weights:

$$\theta_{\text{avg}} = \frac{1}{K + 1} \sum_{k=0}^K \theta'_k. \quad (3)$$

Averaging these parameter sets integrates local optima and encourages convergence toward a stable region in the loss surface, reducing reliance on individual samples. Since adaptation only affects the affine components of the normalization layers, overfitting is mitigated, consistent with the techniques described in Chen et al. (2025). The resulting θ_{avg} , optimized from a single target input, is then reused to initialize the model for subsequent steps, allowing continuous adaptation across the test stream.

3.3.2. Online inference

After completing the adaptation step for the target image volume $\mathbf{x}^t$, the averaged parameters θ_{avg} are frozen and used for final inference. Among the K augmented branches, the one whose prediction achieves the highest Dice similarity with the averaged pseudo-label is selected, and its corresponding query feature statistics are retained. These statistics are reapplied via AFM by aligning them with the encoder output $g(\mathbf{x}^t)$ under the updated parameters θ_{avg} . This reapplication ensures that the final prediction is generated not from the raw target features, but from the statistically adapted representation that most closely aligned with the target distribution during adaptation. The resulting representation is forwarded to the decoder to generate the final segmentation output.

Table 1

Semantic segmentation results (DSC in %, HD in parentheses) on the brain CT dataset for the CTTA adaptation task using ResUNet-50. The best performance is shown in **bold**, and the second-best is underlined.

Domains	Source → Target Domain						Average	
	A → B, C		B → A, C		C → A, B		Eye	Brainstem
Methods	Eye	Brainstem	Eye	Brainstem	Eye	Brainstem		
Testset	85.70(2.39)	83.87(3.22)	83.96(2.53)	83.91(3.34)	54.54(81.68)	77.32(4.97)	74.73(28.87)	81.70(3.84)
w/o Adaptation	67.31(19.18)	70.80(6.55)	45.12(127.58)	56.48(8.64)	47.01(61.93)	70.62(6.65)	53.15(69.56)	65.97(7.28)
TENT (ICLR '21) (Wang et al., 2021)	65.03(35.66)	65.95(23.90)	54.40(15.19)	50.04(10.46)	28.13(172.78)	56.54(15.20)	49.19(74.54)	57.51(11.99)
CoTTA (CVPR '22) (Wang et al., 2022)	56.01(36.56)	58.76(23.90)	67.98(12.96)	<u>65.18(11.38)</u>	23.81(179.57)	53.41(25.21)	49.27(76.36)	59.12(20.16)
MEMO (NIPS '22) (M. Zhang et al., 2022)	60.70(37.02)	66.09(17.17)	74.23(11.49)	27.15(9.05)	22.21(175.80)	63.67(17.24)	52.38(74.77)	52.30(14.49)
SAR (ICLR '23) (Niu et al., 2023)	56.19(42.87)	58.96(27.26)	<u>68.90(12.45)</u>	62.84(11.18)	25.53(177.85)	55.22(24.49)	50.21(77.72)	59.01(20.98)
VPTTA (CVPR '24) (Chen et al., 2024)	<u>67.15(19.21)</u>	<u>70.72(6.58)</u>	45.43(127.07)	56.33(8.50)	<u>46.74(61.82)</u>	<u>70.51(6.68)</u>	<u>53.11(69.37)</u>	<u>65.85(7.25)</u>
FAWA (Ours)	72.86(17.37)	74.20(6.03)	66.97(29.46)	71.61(6.15)	64.53(8.75)	69.50(7.22)	68.12(18.48)	71.77(6.47)

4. Experiments

4.1. Dataset and experimental settings

4.1.1. Dataset

The Brain CT dataset comprises domain A (GLIS-RT Shusharina and Bortfeld, 2021), domain B (Burdenko Zolotova et al., 2023), and domain C (Head-Neck-PET-CT Vallières et al., 2017) with 230, 176, and 44 CT volumes, respectively. Each scan is annotated for the eye and brainstem. Pre-processing included resampling to a voxel spacing of (1.0, 1.0, 1.5), cropping a brain-centred region of interest (ROI) of $256 \times 256 \times 128$ voxels, followed by windowing (W 80, L 40). The Prostate MRI dataset contains domain A (NCI-ISBI13 Bloch et al., 2015), domain B (PROMISE12 Litjens et al., 2014), and domain C (Medical Segmentation Decathlon Simpson et al., 2019) with 78, 50, and 32 MRI volumes, respectively, all annotated for the prostate. Volumes were resampled to isotropic 1.0 and zero-padded to $384 \times 384 \times 40$. Due to its limited size in MRI dataset, domain C was used exclusively for TTA.

4.1.2. Implementation details

In all settings, 10% of each domain was used as the test set. For each adaptation scenario, the model was trained on a single domain and subsequently evaluated on the remaining domains to simulate a continual distribution shift. During pre-training, 3D U-Net with ResNet-50 and ResNet-18 backbones was used for CT and MRI datasets, respectively. All models were optimized using the AdamW optimizer with a batch size of 2 for 200 epochs. The learning rates were set to $1e-3$ for BrainCT and $1e-4$ for ProstateMRI. During TTA, FAWA was applied with a batch size of 1, using AdamW optimizer and learning rates of $1e-4$ and $1e-6$ for BrainCT and ProstateMRI, respectively. For both tasks, the top- K was set to 10, adaptation was performed over $L = 2$ iterations per test sample, and the memory bank capacity was fixed at 30 across all experiments. For a fair comparison, all compared methods were implemented using their official code releases. The learning rate of each method was tuned to adapt it to our segmentation setting, with the configuration yielding the highest average DSC selected as follows: $1e-4$ for TENT and SAR, $1e-5$ for CoTTA, $1e-6$ for MEMO, and $5e-2$ for VPTTA. Performance is reported using the Dice similarity coefficient (DSC) and Hausdorff distance (HD).

4.2. Comparison with state-of-the-art methods

The evaluation compared entropy-minimization methods TENT (Wang et al., 2021) and SAR (Niu et al., 2023) with augmentation-based approaches MEMO (M. Zhang et al., 2022) and CoTTA (Wang et al., 2022), and also included the prompt-based method VPTTA (Chen et al., 2024). The performance range of CTTA was defined by the in-domain test set results (*Testset*) as the upper bound and by the non-adaptive results on unseen domains (*w/o Adaptation*) as the lower bound, serving as the baseline. Full scores for each target domain appear in Table 1 for Brain CT and Table 2 for Prostate MRI. In Brain

CT segmentation, FAWA achieved the highest DSC and the lowest HD for both the eye and brainstem, outperforming the baseline as well as all competing CTTA methods. Several existing approaches even performed worse after adaptation than before, reinforcing earlier findings that weakly supervised objectives such as entropy minimization often struggle with small, detailed anatomy (Chen et al., 2024; Zhang et al., 2024). Specifically, the entropy minimization in TENT and SAR presumes that most pixels belong to confidently identifiable classes, whereas the abundant high-entropy pixels along boundaries and small structures are instead pushed toward over-confident predictions, producing single-class over-segmentation. MEMO was the weakest overall, suggesting that AugMix offers limited benefit in the relatively homogeneous context of three-dimensional medical images. More broadly, the pixel-level augmentations in MEMO and CoTTA disrupt the structural consistency of medical images and can destabilize weight averaging-based adaptation, consistent with the performance drop observed in our ablation (Section 4.3.1). In Prostate MRI, similar patterns appeared. Most methods failed to surpass the baseline, whereas FAWA and the prompt-based VPTTA showed meaningful gains. VPTTA's behavior, however, was dataset-dependent, as freezing the backbone through prompt-tuning avoids collapse but constrains its adaptation capacity, yielding gains on Prostate MRI yet only near-baseline performance on Brain CT. In the setting where part of domain A was used as the source, TENT recorded the highest DSC but a sub-optimal HD, whereas FAWA provided the lowest HD, indicating more precise boundary delineation. Qualitative results in Fig. 3 corroborate the quantitative trends. In CT volumes, CoTTA, MEMO and SAR frequently over-segmented the orbital region, and VPTTA produced fragmented brainstem contours similar to those from the pre-trained model. In MRI volumes, CoTTA, MEMO and SAR often introduced false positives in the prostate region. Across both 3D volumes and representative 2D slices, FAWA produced segmentations that most closely matched the ground truth, confirming its superiority over existing CTTA approaches.

4.3. Impact of components

4.3.1. Ablation study of AFM and CoWA

To assess the contribution of each component, an ablation study was conducted on brain CT segmentation, as presented in Table 3. To evaluate the effect of the adaptively applied mixing parameter in AFM, an additional configuration, referred to as Feature Matching (FM), was constructed by fixing the parameter to a constant value ($\lambda = 0.5$). Both settings were evaluated under the same condition with $K = 10$, where the resulting $K + 1$ predictions were used exclusively for pseudo-label generation. The prediction from the branch exhibiting the highest DSC was subsequently employed in both the adaptation and inference phases. AFM consistently outperformed FM across all three adaptation setups, achieving superior performance in both DSC and HD metrics. On average, AFM improved DSC from 56.32/63.42 to 58.94/64.56 and HD from 43.36/7.95 to 29.64/7.67 for the eye and brainstem, respectively, confirming the benefit of adaptive mixing. However, AFM alone did not

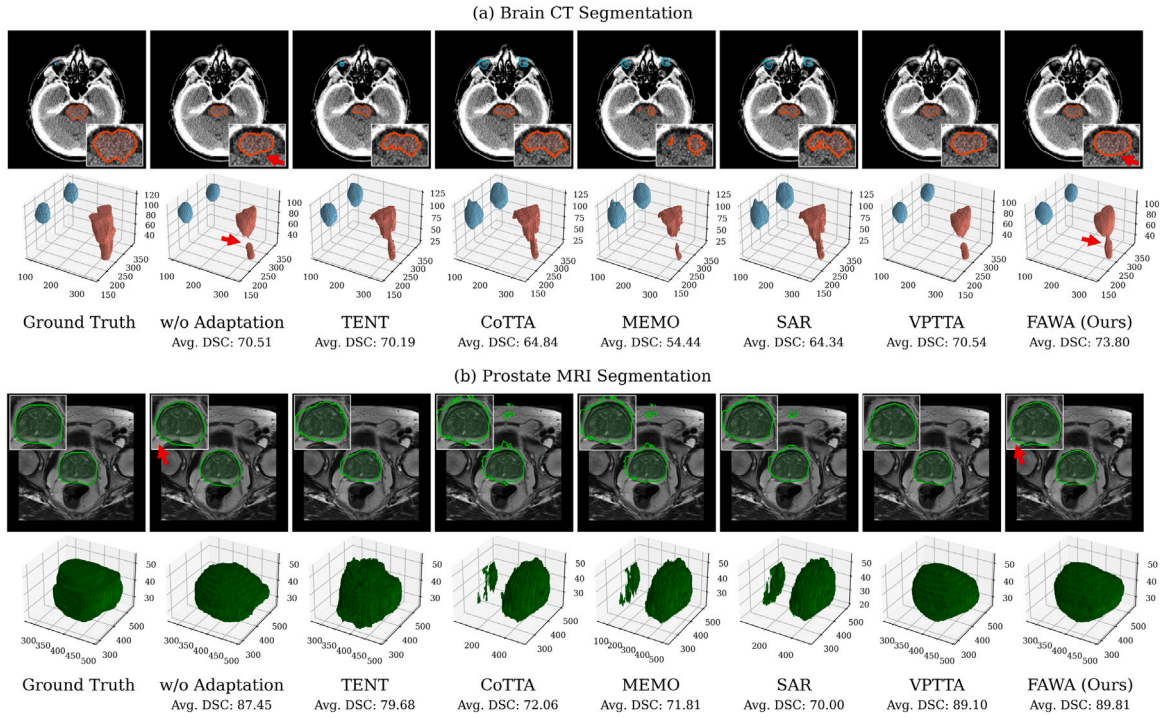


Fig. 3. Visualization of segmentation results for (a) brain CT and (b) prostate MRI. Identical x, y, and z-axis ranges were used for all methods in (a) and for w/o Adaptation, TENT, VPTTA, and FAWA in (b) to ensure a fair 3D comparison.

Table 2

Semantic segmentation results (DSC in %, HD in parentheses) on the prostate MRI dataset for the CTTA adaptation task using ResUNet-18.

Domains	Source $\rightarrow$ Target Domain		Average
	A $\rightarrow$ B, C	B $\rightarrow$ A, C	
Methods	Prostate	Prostate	Prostate
Testset	73.11(10.23)	77.63(4.78)	75.37(7.51)
w/o Adaptation	51.68(12.83)	62.26(12.26)	56.97(12.55)
TENT (ICLR '21)	57.83(53.72)	44.68(41.05)	51.26(47.39)
CoTTA (CVPR '22)	46.11(101.96)	39.56(78.23)	42.84(90.10)
MEMO (NIPS '22)	42.89(68.61)	32.45(45.60)	37.67(57.11)
SAR (ICLR '23)	40.92(99.24)	28.84(113.32)	34.88(106.28)
VPTTA (CVPR '24)	54.06(13.18)	63.23(11.86)	58.65(12.52)
FAWA (Ours)	53.67(12.70)	63.47(11.65)	58.57(12.18)

consistently surpass the baseline in domains A and C, suggesting limited robustness under distribution shifts without additional components. To further assess the contribution of each component, the second module of FAWA, CoWA, was incorporated into the AFM setting. This integration led to substantial improvements across all test conditions, consistently surpassing the baseline. In particular, the average DSC increased to 68.12/71.77 and HD decreased to 18.48/6.47 for the eye and brainstem. These findings indicate that the two components, AFM and CoWA, are complementary and jointly contribute to stable and cumulative gains in segmentation performance. As CoWA relies on a memory bank, its replacement policy was additionally examined by replacing LFU with FIFO and Random under the same protocol with $K = 10$ over the full stream. The Random policy degraded performance substantially, yielding an average DSC of 53.89/64.29 and HD of 52.13/7.65 for the eye and brainstem, respectively, whereas FIFO remained comparable to the adopted LFU policy, attaining an average DSC of 66.56/72.39 and HD of 15.89/6.36. Overall, LFU achieved the most balanced average DSC, supporting its adoption.

CoWA was also benchmarked against conventional pixel-level augmentation by replacing AFM with two image-space strategies, a simple

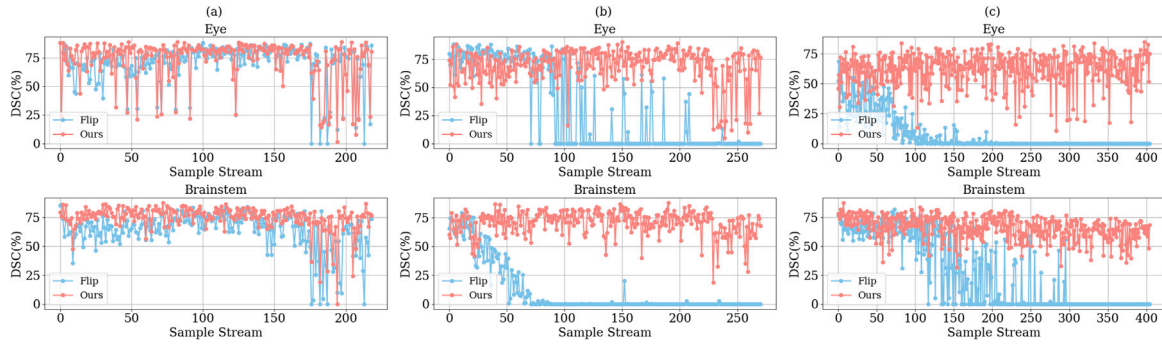
3D flip and SimAug, which applies random cropping, resizing and flipping. In line with the AFM protocol, 10 augmented images were generated per sample. The results show that AFM, which operates in feature space, consistently outperformed both pixel-level schemes. A pronounced performance drop was observed when the source model was trained on domains B and C, with DSC scores dropping to 40.86/35.98 and 27.62/24.61 for the eye and brainstem, respectively. As illustrated in Fig. 4, segmentation accuracy deteriorated sharply after the model processed approximately 100 consecutive test volumes. Without careful tuning, pixel-level augmentation can lead to unstable predictions, resulting in error accumulation and steering the model toward sub-optimal minima. This failure mode is particularly pronounced in weight averaging-based methods, where the effectiveness of parameter aggregation depends critically on how augmented solutions are distributed in the loss landscape. Unlike training-time augmentations, which benefit from sufficient optimization time and multiple iterations, CTTA requires the model to adapt rapidly to each incoming sample. The weight averaging approach further complicates this process, as combining predictions from highly diverse inputs may result in conflicts between diversity and locality (Rame et al., 2022; Zhang et al., 2023). When the weights of augmented samples are too independent, averaging across distant minima in the loss landscape may fail. Conversely, if they are too similar, the lack of diversity may reduce the effectiveness of model updates. This trade-off makes augmentation design even more critical in WA-based methods. In this context, AFM mitigates these challenges by generating feature-level variants that preserve diversity while being guided toward the target distribution and remain within a range that the model can reliably accommodate. As a result, effective adaptation is achieved within a few iterations, without the need for an explicit augmentation search process.

Beyond pixel-level comparisons, we further examined whether the gains of AFM stem specifically from modeling higher-order feature statistics rather than feature-space alignment in general. To this end, AFM was replaced with AdaIN, a representative feature-space method that aligns only mean and variance statistics, while keeping CoWA unchanged and following the same protocol with $K = 10$. As reported

Table 3

Results of ablation study on the brain CT segmentation task (DSC in %, HD in parentheses).

Domains	A $\rightarrow$ B, C		B $\rightarrow$ A, C		C $\rightarrow$ A, B		Average	
Methods	Eye	Brainstem	Eye	Brainstem	Eye	Brainstem	Eye	Brainstem
w/o Adaptation	67.31(19.18)	70.80(6.55)	45.12(127.58)	56.48(8.64)	47.01(61.93)	70.62(6.65)	53.15(69.56)	65.97(7.28)
w/ FM ($\lambda = 0.5$)	62.21(18.49)	61.70(8.06)	57.45(100.86)	64.37(7.54)	49.31(10.73)	64.20(8.26)	56.32(43.36)	63.42(7.95)
w/ AFM	66.35(17.89)	63.07(7.83)	62.43(59.91)	65.74(6.94)	48.03(11.11)	64.86(8.23)	58.94(29.64)	64.56(7.67)
3D Flip + CoWA	70.22(26.18)	64.37(8.68)	45.05(22.02)	10.93(9.91)	7.30(73.20)	32.64(14.48)	40.86(40.47)	35.98(11.02)
SimAug + CoWA	47.65(37.71)	47.86(11.37)	30.50(25.84)	1.23(11.97)	4.71(89.02)	24.73(15.29)	27.62(50.86)	24.61(12.88)
AdaIN + CoWA	71.86(17.38)	74.14(5.85)	66.67(30.31)	70.65(6.51)	63.02(8.91)	65.67(7.91)	67.18(18.86)	70.15(6.75)
AFM + CoWA	72.86(17.24)	74.20(6.03)	66.97(29.46)	71.61(6.15)	64.53(8.75)	69.50(7.22)	68.12(18.48)	71.77(6.47)

**Fig. 4.** Performance comparison of FAWA (AFM+CoWA) and flip+CoWA for Brain CT segmentation in the CTTA setting with continuous input streams. (a), (b), and (c) show the results of adapting the model to the remaining domains when trained on source domains A, B, and C, respectively.**Table 4**

Ablation study on the layer-wise placement of AFM in the adaptation task.

Methods	ResUNet-50		ResUNet-18	Average
Layers	Eye	Brainstem	Prostate	
+ Skip-1	49.77(57.13)	53.56(5.46)	59.26(13.97)	54.20(25.52)
+ Skip-1,2	50.33(63.86)	69.35(6.81)	59.46(13.47)	59.71(28.05)
+ Skip-1,2,3	49.21(62.35)	55.22(5.87)	59.68 (13.30)	54.70(27.17)
+ Skip-1,2,3,4	51.70(63.07)	69.30(7.16)	58.35(13.00)	<u>59.78</u> (27.74)
+ Bottle.	53.13(47.98)	68.71(6.94)	56.91(<u>12.62</u>)	59.58(22.51)
+ Skip + Bottle.	68.12(18.48)	71.77(6.47)	58.57(12.18)	66.15(12.38)

in Table 3, AdaIN+CoWA achieved an average DSC of 67.18/70.15 and HD of 18.86/6.75 for the eye and brainstem, respectively, which is slightly lower than the proposed configuration of AFM+CoWA. This result suggests that aligning only low-order statistics may not fully capture the target feature distribution under continual distribution shifts, whereas AFM accommodates such variations through its adaptive mixing process. Together with the preceding comparisons, these observations support that AFM provides a feature-space alignment that is complementary to the adaptation dynamics enabled by CoWA, contributing to the overall robustness of FAWA.

4.3.2. Performance of AFM across layers

In AFM, the EFDM that directly mixes feature vectors was integrated into the skip connections and bottleneck layers of U-Net-based segmentation models. To quantitatively assess the effect of EFDM placement on model adaptation performance, two configurations were compared using ResUNet-50 and ResUNet-18. The first configuration progressively inserted EFDM starting from the initial skip connection, while the second applied it exclusively at the bottleneck layer. As shown in Table 4, applying EFDM to all skip connections and the bottleneck layer yielded the best performance, achieved an average DSC of 66.15 and HD of 12.38. In contrast, inserting EFDM only into the first four skip layers results in an average DSC of 59.78, while applying it solely at the bottleneck layer achieved an average HD of 22.51. These values represent relative declines of 6.37 and 10.13, respectively, and correspond

to the second best performance in each metric. These results indicate that the effectiveness of EFDM varies with its placement within the network, and that multi-level integration contributes more significantly to adaptation performance.

4.3.3. Effect of CoWA and optimization strategies

To examine the impact of CoWA and its associated optimization strategies on model adaptation, segmentation performance was assessed from multiple perspectives, as shown in Table 5. The weighted averaging strategy used in CoWA, which updates model parameters by aggregating multiple independently fine-tuned weights, was compared against a simplified alternative, referred to as *feature averaging*, in which the mean of the FA feature vectors is computed and only the corresponding weight is updated. Under identical conditions (*full stream*, $K=10$), the weighted averaging approach consistently achieved superior segmentation performance. These results suggest that although *feature averaging* is computationally more efficient, it fails to capture the optimization dynamics inherent to each adaptation path. In contrast, CoWA integrates individually optimized weights from multiple augmented views, enabling the model to combine diverse yet compatible updates in parameter space. This parameter-level aggregation leads to more stable and effective adaptation than averaging features prior to optimization.

Next, model optimization strategies were examined. To address the issue of error accumulation that may arise during CTTA, we compared two distinct strategies: *per sample* and the *full stream* strategy from FAWA. The key difference between these approaches lies in how model updates are managed across samples. After the model is adapted using the i th sample, the $(i+1)$ -th sample can be processed either by resetting the model to its pre-trained state or by retaining the adapted state. The *per sample* strategy adopts the former, enabling the model to adapt independently to each input and thereby reducing the risk of error accumulation. In contrast, the *full stream* strategy maintains the adapted state across samples, promoting cumulative continuous learning. A comparison of the two strategies showed that the *per sample* method surpassed the baseline, while the *full stream* approach led to further improvements, raising scores from 57.54 to 68.12 for the eye, and from 67.80 to 71.77 for the brainstem, respectively. Notably, in the case of

Table 5

Ablation study comparing the performance of the weighted averaging and feature averaging methods under various optimization strategies.

Domains			A $\rightarrow$ B, C		B $\rightarrow$ A, C		C $\rightarrow$ A, B		Average	
Methods			Eye	Brainstem	Eye	Brainstem	Eye	Brainstem	Eye	Brainstem
Feature	Per sample	K=10	68.05(18.58)	72.00(6.04)	47.78(131.56)	56.82(8.31)	52.22(27.36)	69.90(6.89)	56.02(59.17)	66.24(7.08)
Averaging	Full stream	K=10	65.20(19.00)	68.86(6.82)	9.41(102.06)	6.96(3.30)	43.52(12.25)	44.47(11.11)	39.38(44.44)	40.10(7.08)
	Per sample	K=10	69.39(17.99)	72.97(5.93)	50.12(133.67)	59.59(7.91)	53.11(23.82)	70.84(6.66)	57.54(58.49)	67.80(6.83)
Weight Averaging	Full stream	K=1	71.46(17.64)	72.53(6.13)	18.05(145.31)	24.32(8.94)	57.82(10.16)	65.87(7.86)	49.11(57.70)	54.24(7.64)
		K=3	69.78(18.34)	72.82(6.15)	39.12(122.01)	57.37(8.51)	60.20(9.63)	70.04(7.03)	56.37(49.99)	66.74(7.23)
		K=5	71.07(17.79)	74.30(6.14)	67.98(19.29)	<u>70.42(6.38)</u>	59.76(9.82)	69.40(7.15)	<u>66.27(15.63)</u>	<u>71.37(6.56)</u>
		K=10	72.86(17.24)	74.20(6.03)	<u>66.97(29.46)</u>	71.61(6.15)	64.53(8.75)	69.50(7.22)	68.12(18.48)	71.77(6.47)
		K=20	72.46(18.10)	74.60(6.05)	<u>60.26(22.66)</u>	63.14(7.74)	56.61(9.68)	60.55(8.67)	63.11(16.81)	66.10(6.90)
		K=30	72.28(18.33)	74.80(6.03)	51.23(29.27)	53.74(9.27)	58.83(9.76)	62.83(8.43)	60.78(19.12)	63.79(7.65)

Table 6Semantic segmentation results (DSC in %) on the Brain CT dataset over three rounds of long-term CTTA. The performance gap (Δ Perf.) is computed as the difference between the average DSC in each round and the baseline.

Time		$t \longrightarrow$															Times
Round		1					2					3					
Methods	Domains	A	B	C	Avg.	Δ Perf.	A	B	C	Avg.	Δ Perf.	A	B	C	Avg.	Δ Perf.	
w/o Adaptation	Eye	67.31	45.12	47.01	53.15	–	67.31	45.12	47.01	53.15	–	67.31	45.12	47.01	53.15	–	–
	Brainstem	70.80	56.48	70.62	65.97	–	70.80	56.48	70.62	65.97	–	70.80	56.48	70.62	65.97	–	
TENT	Eye	65.03	54.40	28.13	49.19	–3.96	40.06	0.77	2.38	14.40	–38.75	0.64	0.01	0.01	0.22	–52.93	0.748
	Brainstem	65.95	50.04	56.54	57.51	–8.46	74.94	0.13	0.03	25.03	–40.94	69.24	1.73	0.03	23.66	–42.30	
CoTTA	Eye	56.01	67.98	23.81	49.27	–3.88	54.83	65.70	14.45	44.99	–8.16	53.74	64.28	2.60	40.21	–12.94	4.085
	Brainstem	58.76	65.18	53.41	59.12	–6.85	57.63	68.13	51.22	58.99	–6.98	56.89	67.53	52.49	58.97	–7.00	
MEMO	Eye	60.70	74.23	22.21	52.38	–0.77	72.21	75.10	3.83	50.38	–2.77	75.14	67.91	0.03	47.69	–5.46	6.988
	Brainstem	66.09	27.15	63.67	52.30	–13.67	61.12	27.04	66.47	51.54	–14.43	13.26	1.17	27.55	13.99	–51.98	
SAR	Eye	56.19	68.90	25.53	50.21	–2.94	56.18	68.89	25.52	50.20	–2.95	56.18	68.89	25.52	50.20	–2.95	1.591
	Brainstem	58.96	62.84	55.22	59.01	–6.96	58.96	62.86	55.22	59.01	–6.96	58.96	62.86	55.22	59.01	–6.96	
VPTTA	Eye	67.15	45.43	46.75	53.11	–0.04	67.15	45.47	46.75	53.12	–0.03	67.14	45.49	46.75	53.13	–0.02	0.583
	Brainstem	70.72	56.33	70.51	65.85	–0.12	70.72	56.33	70.51	65.85	–0.12	70.71	56.33	70.51	65.85	–0.12	
FAWA (Ours)	Eye	72.86	66.97	64.53	68.12	+14.97	73.08	67.72	67.70	69.50	+16.35	72.58	68.16	68.15	69.63	+16.48	18.36
	Brainstem	74.20	71.61	69.50	71.77	+5.80	74.34	70.11	66.39	70.24	+4.27	73.16	68.66	61.66	67.83	+1.86	

feature averaging, the *per sample* strategy yielded better performance than *full stream*. This can be explained by the compounding effect of information loss caused by feature averaging, which led to accumulated errors in the *full stream* setting. These findings confirm the effectiveness of CoWA in enhancing performance and highlight the critical role of optimization strategies in CTTA. In particular, leveraging accumulated information in CoWA enables more effective model adaptation to subsequent samples.

Finally, performance variation with respect to the FA retrieval parameter K , a core component of CoWA, was analyzed under the *full stream* setting. Results showed that even at $K = 3$, CoWA outperformed the baseline. As K increased to 5 and 10, average DSC and HD scores for both the eye and brainstem classes continued to improve. However, further increasing K to 20 and 30 led to performance degradation. This trend aligns with the observation that cross-domain shape similarity is closely associated with transfer performance (Zhang et al., 2024). When using cosine similarity for top- K retrieval, larger K values may include semantically less relevant samples, ultimately harming adaptation effectiveness. Overall, the best performance was achieved at $K = 10$.

4.4. Additional analyses

4.4.1. Multiple rounds of TTA

An extended component analysis of FAWA was conducted under a CTTA setting to evaluate model robustness over multiple adaptation rounds, simulating intensified and prolonged domain shifts. The experiment included up to three sequential adaptation steps per domain using a brain CT segmentation model, alongside comparisons with SOTA methods. For each round, DSC values were recorded and compared to the baseline, with the difference denoted as Δ Perf., allowing analysis of performance trajectories across iterations. As illustrated in

Table 6, both TENT and MEMO showed notable degradation after the second round, highlighting the limitations of entropy-based adaptation. While entropy minimization can yield competitive performance in early adaptation steps, it is prone to degenerate solutions in which the objective is satisfied by confidently predicting a dominant class regardless of input content, leading to error accumulation and eventual performance collapse under prolonged continual adaptation. Although CoTTA maintained relatively stable segmentation performance for the brainstem class, its performance declined substantially in the eye class, resulting in inconsistent adaptation across anatomical regions. VPTTA, which relies on low-frequency prompt tuning rather than direct parameter updates, maintained moderate accuracy but failed to yield further gains over time. SAR exhibited some resistance to error accumulation through spatially adaptive normalization, but its overall accuracy remained considerably lower than that of FAWA. In contrast, FAWA demonstrated consistent performance across adaptation rounds. Although a minor decline was observed in the brainstem class, its performance remained above the baseline throughout. Notably, for the eye class, the average DSC improved steadily from 68.12 in round 1 to 69.63 in round 3, indicating that FAWA successfully adapted over time without exhibiting error accumulation.

4.4.2. Revisiting source distribution for catastrophic forgetting

Continual adaptation over extended target streams inevitably modifies the affine parameters toward the target distribution, leading to progressive degradation in the source domain. To quantify this, the adapted model was saved at the end of each round and re-evaluated on the source test set. As shown in Table 7, the gaps in DSC (Δ CF_{DSC}) and HD (Δ CF_{HD}) relative to the pre-trained baseline confirm a consistent decline across rounds, with the sharpest drop occurring between rounds 2 and 3, an expected consequence of parameter shift under more than 400 volumetric cases per round. The more notable

Table 7

Performance comparison after revisiting the source domain using adapted models following three rounds of CTTA.

Methods		Eye				Brainstem			
		DSC _{Avg.} ↑	HD _{Avg.} ↓	Δ CF _{DSC} ↑	Δ CF _{HD} ↓	DSC _{Avg.} ↑	HD _{Avg.} ↓	Δ CF _{DSC} ↑	Δ CF _{HD} ↓
Pre-trained Model	Testset	74.73	28.87	–	–	80.84	3.84	–	–
After Adaptation	Round 1	74.47	16.85	–0.26	–12.02	74.95	5.46	–5.89	1.62
	Round 2	70.85	17.34	–3.88	–11.53	65.69	7.94	–15.15	4.10
	Round 3	52.71	26.11	–22.02	–2.76	55.50	12.45	–25.34	8.61
TTA w/ Adapted Model	Round 1	75.17	16.82	0.44	–12.05	76.21	5.23	–4.63	1.39
	Round 2	72.55	17.19	–2.18	–11.68	69.97	7.20	–10.87	3.36
	Round 3	62.59	18.86	–12.14	–10.01	57.12	13.49	–23.72	9.65

Table 8Ablation study on the effect of the iteration hyperparameter L during sample-wise adaptation on the prostate MRI dataset.

Domains	A → B, C	B → A, C	Average	Times (s/case)
Iteration 1	53.41(11.70)	63.28(11.81)	58.35(11.76)	7.807
Iteration 2	53.67(12.70)	63.47(11.65)	58.57(12.18)	8.612
Iteration 3	53.59(13.96)	63.29(11.61)	58.44(12.79)	12.037

finding, however, is that applying FAWA at inference on the source domain using the adapted model (*TTA w/ Adapted Model*) substantially recovers this degradation across all classes and rounds. In round 3, for example, the DSC of the eye class rose from 52.71 to 62.59, an absolute gain of nearly ten percentage points. In round 1, performance even exceeded the pre-trained baseline, with DSC and HD improving from 74.73/28.87 to 75.17/16.82, reducing the CF gap to Δ CF_{DSC} 0.44 and Δ CF_{HD} –12.05. These results suggest that FAWA's memory-guided feature mixing compensates for the distributional gap introduced by the parameter shift, demonstrating that the framework retains generalization capability beyond the target stream without revisiting the original training data.

4.4.3. Analysis of hyperparameter selection

The influence of the iteration hyperparameter L is summarized in Table 8. In FAWA, a single iteration comprises independent fine-tuning on both the original and an augmented view of each input, followed by weight averaging. Experiments evaluated $L \in \{1, 2, 3\}$. Optimal performance was achieved at $L = 2$, yielding DSC scores of 53.67 and 63.47 for the two prostate adaptation settings, respectively. In contrast, setting L to 1 or 3 degraded performance, resulting in lower average DSC values of 58.35 and 58.44. These results accord with previous work (Zhang et al., 2023), which shows that the benefit of weight averaging depends on the balance between optimization depth and averaging frequency. A moderate number of FA iterations broadens the diversity of local optima relative to a single step, however, further iterations can drive individual solutions beyond the loss basin's vicinity, weakening the averaging effect and eroding the gains of FAWA. In particular, even at $L=1$, FAWA shows an improvement over the non-adaptive baseline, suggesting that the number of iterations can be minimized to reduce inference latency without sacrificing robustness. Thus, although the choice of L is necessary for optimal performance, the evidence indicates that a small set of parallel adaptation steps followed by model averaging reliably improves performance in online CTTA settings.

5. Limitation

Despite its robust performance, FAWA entails a higher computational overhead compared to existing TTA approaches. As shown in Tables 6 and 8, the inference time per case is 18.36 s for Brain CT and 8.612 s for Prostate MRI, compared to 0.583 s for VPTA and 6.988 s for MEMO in Brain CT. This overhead stems from FAWA's parallel branch structure, which independently fine-tunes $K + 1$ parameter sets per sample before averaging. The cost, however, is tunable rather

than fixed. Reducing K from 10 to 5 lowers the inference time to 17.48 s while retaining the second-best accuracy across all K values (eye 66.27/brainstem 71.37, see Table 5), and $K = 3$ further reduces it to 15.68 s while still surpassing the non-adaptive baseline. Similarly, as shown in Table 8, decreasing the number of iterations L provides an additional axis for controlling computational cost with minimal performance degradation. These operating points allow practitioners to navigate the runtime-accuracy trade-off according to deployment constraints. For per-case CTTA workflows in which adaptation is followed by clinician reading, the overhead is comfortably absorbed within the existing acquisition-to-report timeline. In contrast, for latency-critical settings, a smaller (K, L) configuration provides a faster alternative with bounded accuracy loss. Further reducing inference latency while preserving adaptation gains is an important direction for future work.

6. Conclusion

This study shows that pixel-level augmentation, common in many CTTA pipelines, can amplify prediction errors during online medical image segmentation. To overcome this issue, we introduce FAWA, an augmentation strategy that operates entirely in feature space and employs continual weight averaging for stable one-pass adaptation. On Brain CT and Prostate MRI streams, FAWA achieved competitive performance against existing CTTA methods and maintained its advantage over multiple adaptation rounds, demonstrating strong robustness in realistic clinical scenarios. These results indicate that FAWA can reduce inter-institutional variability and improve segmentation accuracy under realistic distribution shifts.

CRedit authorship contribution statement

Kyuri Kim: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Hyungjoo Cho:** Software. **Yoonho Na:** Visualization, Validation, Methodology, Investigation. **Jimin Lee:** Writing – review & editing, Investigation. **Sung-Joon Ye:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

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Data availability

Data will be made available on request.

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